

MODEL QUESTION PAPER - 1
Ground Improvement Techniques

Time: 3 Hour

Max.Marks: 75

PART-A

I. Answer all questions in one word or one sentence

(9 x 1 = 9 Marks)

Sl. No	Question
1	Name the type of roller which uses compressed air to develop inflation pressure for compaction of soil.
2	The method of dewatering by electro osmosis is most suitable for which type of soil.
3	A method for improving clay formations by accelerating consolidation
4	The continuous drainage constructed beneath dams and basements floor slabs to provide highly permeable drainage path
5	Injection of fluidized material into the voids of the ground of spaces between the ground and adjacent structures
6	The most commonly used geosynthetic for reinforced earth walls are
7	Define compressibility of soil
8	Name any one raw material used for manufacture of geosynthetics.
9	The ratio of D_{15} to D_{85} is termed as _____

PART-B

II. Answer any eight questions from the following. Each question carries 3 marks.

(8 x 3 = 24 Marks)

1	Explain the method of standard proctor test for determination of Optimum moisture content
2	List the different methods of de-watering systems.
3	Illustrate a Stone column and its installation
4	How is single stage well point system different from multi stage well point system.
5	Discuss the types of grouts used for ground improvement techniques.
6	Discuss the three basic functions involved in soil or rock grouting.

7	List the different methods of soil stabilization
8	List the different types of geosynthetics used for ground improvement.
9	Write brief note on compaction grouting.
10	Discuss the working principle of vertical drains. Also name the two types of vertical drains.

PART-C

III. Answer ALL questions. Each question carries 7 marks.

(6 x 7 = 42 Marks)

III	Explain the dewatering method by electro osmosis
OR	
IV	Explain the working of sheep foot rollers for densification of soil. Also list out the advantages and disadvantages when compared to other rollers.
V	Discuss the stabilization of soil using cement
OR	
VI	Discuss in detail about permeation grouting technique.
VII	Compare and contrast consolidation and compaction
OR	
VIII	Explain the laboratory test procedure for determining the consolidation characteristics of soil
IX	Discuss the different types of reinforcing elements used in reinforced earth structures.
OR	
X	Explain the sequence of construction of a reinforced earth wall? Also list out the causes of failure of reinforced earth walls.
XI	Explain the soil reinforcement mechanism
OR	
XII	Discuss any four engineering applications of geomembrane used as a containment or barrier.
XIII	Explain the preloading technique for soil improvement
OR	
XIV	Discuss the salient features of sand drains

MODEL QUESTION PAPER - 2
Ground Improvement Techniques

Time: 3 Hour

Max.Marks: 75

1. **PART-A**2. **I. Answer all questions in one word or one sentence****(9 x 1 = 9 Marks)**

Sl. No	Question
1	Define Compactive effort in field conditions
2	Use of stone columns is an effective technique for improving _____ soils.
3	The most commonly used geosynthetic for reinforced earth walls are
4	The soil-spring analogy model for consolidation of soil was proposed by
5	The water content corresponding to the peak of a compaction curve is termed as
6	Define Groutability ratio
7	Vacuum dewatering systems are suitable for improving which type of soil
8	The main function that a geosynthetic serves when used in drains
9	Define consolidation

3.

4. **PART-B**5. **II. Answer any eight questions from the following. Each question carries 3 marks.****6. (8 x 3 = 24 Marks)**

1	List the factors affecting the selection of ground improvement technique.
2	Explain the electro-osmosis method of de-watering.
3	List the different rollers used for surface compaction.
4	Explain the open sump dewatering system
5	Discuss the types of grouts used for ground improvement techniques.
6	List the objectives of grouting.
7	Explain the conventional sequence of operation for construction of soil-asphalt.
8	List any three methods of soil stabilization
9	List the five basic functions of geosynthetics
10	Discuss the different methods of preloading.

7. PART-C

8. III. Answer ALL questions. Each question carries 7 marks.

(6 x 7 = 42

Marks)

III	Explain the vacuum dewatering method of ground improvement.
OR	
IV	Explain the working of a stone column. Also list the types of soils that can be improved using the technique.
V	Explain the principle behind the technique of grouting.
OR	
VI	Explain in detail bituminous stabilization of soil
VII	Describe the procedure for finding the coefficient of consolidation in the laboratory
OR	
VIII	Discuss the working principle of vertical drains. Also, explain prefabricated vertical drains.
IX	Describe the differences between geotextiles and geomembranes.
OR	
X	What do you understand by ‘Reinforced Earth’? Discuss the different components of a reinforced earth wall?
XI	Discuss any three different types of geosynthetics used for ground improvement.
OR	
XII	Discuss the functions of geosynthetics when used on pavements.
XIII	Discuss Terzaghi’s soil-spring analogy model for explaining consolidation.
OR	
XIV	Differentiate between consolidation and compaction